

Bitcode

Market Analysis — Research Brief

A two-sided marketplace where code and data are measured, commoditized, bought, and sold — connecting AI model builders (buyers) with IP-holders monetizing code (sellers).

Who uses Bitcode? How much do they use it? How much could it be used?

Field	Detail
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Confidence	14 of 15 load-bearing figures independently confirmed; figures are directional where noted

Prepared as an internal analytical brief. Market-size figures are drawn from third-party analyst reports and reported deal values; they vary by definition and should be read as directional. Confidence levels and caveats are stated throughout.

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1. Executive summary

Bitcode addresses a real and fast-forming market: the acquisition of high-quality data — and code in particular — to train, tune, and evaluate AI models. The core AI training-dataset market is roughly **\$3.9 billion in 2026** and is projected to reach **\$16–\$23 billion by 2033–2034** (Grand View Research and Fortune Business Insights, ~22–23% CAGR). The broader *data-acquisition wallet* — datasets plus labeling, licensing, and managed expert-data services — is already on the order of **\$8–\$12 billion in 2026** and is compounding at 20%+.

Three structural forces make code-as-training-data a compelling wedge: **(1) scarcity** — Epoch AI projects the stock of public human-generated text will be exhausted between 2026 and 2032 (median ~2028); **(2) disproportionate value** — Cohere's research shows code in pre-training improves non-coding reasoning and world-knowledge tasks, and software development is the “killer” enterprise AI use case; and **(3) legal risk** — the dominant code corpora were scraped without clear rights, so rights-cleared, provenance-tracked code is now a competitive necessity, not a nicety.

The wedge. Every major code corpus in use today (The Stack v1/v2, StarCoder2, CodeParrot) is scraped, opt-out-based, and legally contested — the GitHub Copilot suit remains live on open-source-license-violation and breach-of-contract claims, and Anthropic paid **\$1.5 billion** to settle authors' copyright claims (the largest copyright recovery in U.S. history). Bitcode's defensible position is cleanly-licensed, consented, provenance-tracked code with a rights trail buyers can stand behind.

Headline takeaways

- **Concentrated demand.** ~10–15 frontier labs control the vast majority of data spend (OpenAI, Anthropic, Google, Meta, Amazon, Microsoft, xAI, plus China's DeepSeek/Alibaba and Europe's Mistral/Cohere), widening to dozens of model-building and fine-tuning firms.
- **A public price ladder now exists.** Premier news ~\$50M/yr (News Corp–OpenAI, >\$250M/5yr), community/real-time data ~\$60–\$70M/yr (Reddit–Google/OpenAI), mid-tier news \$5–\$25M/yr, stock media \$25–\$50M/deal, academic \$10–\$23M. These set anchors for what buyers will pay for rights-cleared corpora.
- **Sellers are three distinct pools.** Positive-ROI recycled code (legacy / non-moat), dead code (defunct / pivoted companies), and purpose-built “research-for-cash” supply — the last is the fastest-growing, with vendors like Surge (~\$1.2B rev), Scale (\$29B val), Mercor (\$10B val), Turing, Handshake and Invisible now a multi-billion-dollar layer.
- **Take-rate is the key business-model lever.** Passive data marketplaces price at 0–3% (Databricks 0%, AWS Data Exchange 3%, Ocean ~0.3%); curated/verified vendors capture ~30–40% (Mercor) to ~50% (Scale-style markup). Bitcode's achievable rate depends on how much rights-clearing, curation and verification it owns.
- **Sizing the prize.** A code-specific, rights-cleared serviceable market (SAM) of roughly \$1.5–\$3B in 2026 (code as ~15–25% of the data wallet) scaling to \$6–\$12B+ by the early 2030s. At a curated 20–30% take on realistically intermediated GMV, that supports a path from tens of millions to \$100M+ in marketplace revenue.
- **The load-bearing caveat.** Median-enterprise fine-tuning demand has softened as base models improve (a16z); the durable demand is in high-value, verticalized, code/RL-grade data. Bitcode should index to premium, hard-to-synthesize, rights-clean code — not commodity fine-tuning volume.

2. Why now — the market thesis

Four trends converge to make a code/data marketplace timely. Each is independently sourced and, where central, adversarially verified.

2.1 The data wall is arriving

Epoch AI projects, with ~80% confidence, that the stock of publicly available human-generated text will be fully utilized between **2026 and 2032 (median ~2028)**; the indexed web holds ~500 trillion words and is growing far slower than frontier models consume it. Among Epoch's explicitly named mitigations for the data wall is “adding more code data.” As raw web text is exhausted, curated, rights-clean, and novel code becomes a premium input. *Epoch AI (2024–26); confirmed via AP/PBS.*

2.2 Code is disproportionately valuable training data

Cohere For AI's study “To Code, or Not To Code?” found that including code in pre-training improves performance on *non-coding* tasks: up to +8.2% on natural-language reasoning, +4.2% on world knowledge, +6.6% on generative win-rate (and ~12× on code itself), with gains rising with model size. Independent work links code's logical structure and verifiability to better reasoning and entity-tracking. Practically, software development is the leading enterprise AI use case — one CIO survey reports ~90% of some teams' code is now AI-generated — and repository-scale coding tasks and RL environments have become a stated “strategic priority” for frontier labs. *Cohere/VentureBeat (2024); arXiv 2509.21499, 2405.21068; a16z (2025); Sapphire Ventures (2025).*

2.3 Legal risk has repriced “free” data

The dominant code corpora were scraped without clear rights. The GitHub Copilot litigation (Doe v. GitHub, filed Nov 2022) survives on open-source-license-violation and breach-of-contract claims after copyright/DMCA counts were dismissed by 2024. Anthropic agreed to pay **\$1.5 billion** (~\$3,000/book across ~500,000 works) to settle Bartz v. Anthropic — the largest copyright recovery in U.S. history. Microsoft's new MAI models are marketed as trained “on clean, commercially licensed data — no distillation.” Provenance and licensing have become purchasing criteria. *githubcopilotlitigation.com; NPR/Authors Guild (2025); Microsoft AI (2026).*

2.4 The supply side is professionalizing

A purpose-built “research-for-cash” layer now manufactures training and evaluation data at scale, and buyers are integrating it (Meta's \$14.3B for 49% of Scale AI). Pricing is shifting from flat annual fees toward usage-based and revenue-share models. Data-labeling spend has grown far faster than compute (one 2023–24 estimate: labeling +88× vs compute +1.3×), and by some estimates high-quality human data now rivals or exceeds the marginal compute cost of frontier training. Data has moved from an afterthought to a contested, priced asset — exactly the condition a marketplace monetizes. *Time/Forbes (2025); Quartz (2026); D. Kang, UIUC (2025).*

3. Question 1 — Who uses Bitcode?

Bitcode is a two-sided marketplace. On the demand side are AI model builders who need code/data; on the supply side are IP-holders who can monetize code they own. Below, each segment from the deck is defined, sized in count, and mapped to named accounts.

3.1 Buyers — enterprises developing models

Segment A — Frontier Labs

The genuinely frontier-scale buyer set is small and concentrated: roughly **10–15 well-resourced labs** globally. Epoch AI tracked 149 frontier models (end-2021 to end-2025) across ~18 countries (43 U.S., 19 China, 6 U.K.), and projects models exceeding 1e26 FLOP rising from ~10 (2026) to ~80 (2028) to 200+ (2030) — i.e., the buyer set broadens over time. These accounts have the largest data budgets and set the price ladder. *Epoch AI (2025–26); arXiv 2504.16138.*

Target accounts — frontier labs (data-buying power).

Lab	Scale (val / revenue)	Data-spend signal
OpenAI	~\$852B val; ~\$25B ARR	“Hundreds of millions” on licensing (News Corp \$250M/5yr, Reddit ~\$70M/yr, FT \$5–10M/yr, Axel Springer ~\$13M/yr, Dotdash ≥\$16M, Shutterstock \$25–50M) + heavy RLHF/expert data. Prime account.
Anthropic	~\$965B val; ~\$47B ARR; IPO-filed	\$1.5B author settlement = motivated licensed-data buyer; growing licensing/labeling. Coding-leader positioning drives code-data appetite.
Google DeepMind	Alphabet \$175–185B '26 capex	Reddit ~\$60M/yr; huge first-party data (Search/YouTube) reduces external need but sets a high internal bar.
Meta AI	\$66–72B '25 capex	Bought the supplier: \$14.3B for 49% of Scale AI to secure specialized training datasets. Vertically integrating data.
xAI	~\$230B val	Leans on X first-party corpus + synthetic; lower external licensing need, but large model cadence.
Microsoft (MAI)	FY26 capex ~\$100B+	MAI models trained on “commercially licensed data” by design; Informa/T&F \$10M deal. A clean-data buyer.
Amazon (Nova)	up to \$33B into Anthropic	NYT \$20–25M/yr (first publisher deal) + Nova labeling; first-party retail/Alexa data. Expect more deals.
DeepSeek	~\$45–50B val (reported)	Efficiency-first (~\$6M model runs); heavy synthetic/distillation; low disclosed external data spend.
Alibaba (Qwen)	≥\$53B/3yr cloud+AI	Large but opaque; China-sourced data. Epoch flags it as under-covered.
Mistral / Cohere / AI21 / Reka	\$1–20B val range	Smaller but growing buyers; multilingual/enterprise/private-data focus. Acquisition targets as much as big spenders.

Figures for 2026 valuations/ARR draw partly on secondary trackers (Sacra, getlatka) and are directional; deal values, capex guidance and the Scale/Anthropic figures are high-confidence.

Segment B — Other model-building companies

Beyond the frontier, a broad tier builds notable or vertical models: NVIDIA (open Nemotron models *with* open data), Databricks (Mosaic AI), Reka, Contextual AI, and national/vertical model builders. Epoch's trajectory implies **30+ organizations** training notable models by 2027. Many lack first-party data moats and are the natural mid-market for a code/data marketplace.

Segment C — Model-tuning / fine-tuning companies

The tuning layer turns base models into custom, domain-specific systems — and is the segment most dependent on *proprietary and code* data. The custom-LLM-training-platform market is ~\$3.52B in 2026 → \$8.63B by 2030 (~25% CAGR); the broader enterprise-LLM market is ~\$5.65B (2025) → \$41.6B by 2033 (28.3%), with domain-specific LLMs the fastest-growing sub-segment (>38% CAGR). Snorkel's pivot to “Expert Data-as-a-Service” is a direct market signal. *Research and Markets (2026); Grand View Research (2025).*

Target accounts — model-building / tuning ecosystem (buyers of code/domain data).

Company	Funding / valuation	Positioning & data signal
Scale AI	\$29B (Meta 49%)	Incumbent data + RLHF + customization vendor; ~\$870M→\$2B rev. Competitor and proof of market scale.
Snorkel AI	\$1.3B (Series D)	Pivoted to “Expert Data-as-a-Service” for fine-tuning in software dev, healthcare, finance — closest analog/partner.
Databricks (Mosaic AI)	\$134B (Dec 2025)	Owns customers' proprietary data lakes; prime channel and buyer for code + domain data.
Together AI	\$8.3B (Jul 2026)	Train/fine-tune/deploy open models; bought Refuel.ai for data structuring; customers Cursor/Cognition need code data.
Fireworks AI	\$4B (→~\$15B talks)	Inference + fine-tuning for 10,000+ companies; high-volume tuning → downstream data demand.
Cohere	\$7B	Enterprise LLMs; ~85% private deployments on regulated-enterprise proprietary data; IPO-track.
Hugging Face	~\$13.5B	Model + dataset hub with AutoTrain; already hosts a datasets marketplace — key partner/competitor.
Contextual AI / Lamini / H2O.ai	\$0.1–0.6B	Theses built on tuning models with customers' proprietary data (incl. private codebases), often on-prem.
Baseten / Modal	\$4.6–13B	Serverless training/inference infra for teams tuning on proprietary data — volume indicators.

3.2 Sellers — IP-holders monetizing code

The deck's three seller categories map to three distinct supply pools with very different liquidity and pricing today.

Segment D — Positive-ROI recycled data (legacy / non-moat code)

Existing code that is safe to expose (low risk) and profitable to license (high reward): legacy systems, non-differentiating internal code, deprecated products. The raw supply is enormous — GitHub hosts **180M+ developers and ~630M repositories** (Octoverse 2025); Software Heritage archives 27B+ unique source files (2+ petabytes). The gap is not volume but *rights and provenance*: almost none of this is available as cleanly-licensed, sale-ready corpora. This is Bitcode's largest addressable pool. *GitHub Octoverse 2025; Software Heritage (2026)*.

Segment E — Dead code (defunct / pivoted companies)

Code and IP from failed or pivoted companies — “trash for one, treasure for another.” Today this is monetized almost exclusively as *patents* via bankruptcy auctions (Nortel: ~6,000 patents for \$4.5B in 2011; Kodak: ~1,100 for \$525M in 2013), not as source code. The reality is low liquidity and weak price discovery: distressed IP sells at a fraction of value, the brokered patent market was only ~\$290M/yr (2020), and most failed-startup code is never monetized because creditors lack the means or interest. **This is the most under-exploited, highest-latent-supply pool** — and the one where a marketplace with price discovery adds the most value. *Fox Business/Wikipedia (2011–13); SimpleClosure (2024)*.

Segment F — IP made to be sold (research-for-cash)

Purpose-built supply: experts (including coders) paid to produce training, evaluation, and RL data. This is the fastest-growing seller layer and the clearest proof of willingness-to-pay.

Research-for-cash / expert-data vendors.

Vendor	Revenue	Valuation	What they pay contributors (code emphasis)
Surge AI	~\$1.2B (2024)	~\$15–25B (raising)	\$35–\$50/hr RLHF; \$85–\$120/hr specialized coding (Rust, C++). Bootstrapped; ~12 frontier-lab clients.
Scale AI	\$870M→\$2B	\$29B (Meta)	Per-task markup, ~50%+ gross margin; historical piece rates as low as ~2–6¢/task via Remotasks.
Mercor	~\$450M ARR (Oct '25)	\$10B (→\$20B talks)	\$85+/hr avg; ~\$40–\$55/hr RLHF; ~30–40% platform take (contractors keep 60–70%).
Turing	~\$300M ARR	\$2.2B	Supplies coding/SWE data to OpenAI/Google/Anthropic/Meta from 4M+ vetted engineers; profitable.
Handshake AI	~\$1.1B run-rate	\$3.5B+	\$40–\$125/hr (\$100–125 for CS/math/physics); 500k PhD + 3M master's pool via .edu.
Invisible Tech	\$134M (2024)	\$2B+	RLHF pioneer (OpenAI 2022 onward); Amazon, Microsoft, Cohere clients.

Read-through for supply. Coding/software-engineering data already commands the top of the pay scale (\$85–\$125/hr) and is where synthetic-plus-expert generation is a distinct, premium sub-market. A marketplace that can aggregate rights-clean legacy code (Segment D) and unlock dead code (Segment E) taps supply that today is either unavailable or trapped.

4. Question 2 — How much do they use it?

Two reads: a top-down market size, and a bottom-up tally of what buyers actually pay (capex context, the licensing deal ladder, and managed-data-vendor revenues as a proxy).

4.1 Top-down market size

Estimates for the AI training-dataset market cluster tightly for 2026 at the “dataset-only” definition and spread at the out-years by CAGR and scope. Adjacent markets (labeling, licensing, custom-LLM platforms, data marketplaces) bound the broader wallet.

Market (definition)	2025	Out-year	CAGR	Source / confidence
AI training dataset (narrow)	\$3.2B	\$16.3B (2033)	22.6%	Grand View Research — high
AI training dataset (narrow)	\$3.59B	\$23.2B (2034)	22.9%	Fortune Business Insights — high
AI training dataset (conservative)	\$3.35B	\$14.9B (2035)	16.1%	Precedence Research — high
AI training dataset (broad, incl. tools/services)	—	\$52.4B (2035)	24.2%	Business Research Insights — outlier high
Dataset licensing for AI training	\$4.8B	\$22.6B (2034)	18.8%	MarketIntelto — med (thin sourcing)
Data collection & labeling	\$3.77B ('24)	\$17.1B (2030)	28.4%	Grand View Research — high
Custom LLM training platforms	\$2.82B	\$8.63B (2030)	25.1%	Research and Markets — confirmed
Enterprise LLM	\$5.65B	\$41.6B (2033)	28.3%	Grand View Research — high
Data marketplace platforms (comp)	\$1.86B	\$5.73B (2030)	25.2%	Grand View Research — high

Why the spread? The 2026 base is ~\$3.9B where firms count datasets only (GVR \$3.9B, Precedence \$3.92B, FBI \$4.44B) and ~\$7.5B where they fold in collection tools, labeling software, validation services and dataset marketplaces (Business Research Insights). The out-year range (\$13B–\$52B) is driven mostly by CAGR assumptions (16% vs 24%+) and end-year (2033 vs 2035), not disagreement on direction. Verification note: the 22–24% CAGR belongs to GVR and FBI specifically; Precedence models a materially lower ~16% path.

4.2 The data-licensing deal ladder

Disclosed deals reveal what buyers pay for rights-cleared corpora — the most relevant anchor for Bitcode's pricing. 90+ licensing deals have been publicly disclosed; a representative ladder:

Deal	Reported value	Notes
News Corp – OpenAI	>\$250M / 5yr	Largest disclosed; ~\$50M/yr, blends cash + technology credits (“potentially” >\$250M).
Reddit – Google	~\$60M / yr	Real-time community data for Gemini (Feb 2024).
Reddit – OpenAI	~\$70M / yr (est.)	Part of Reddit's \$203M aggregate disclosed licensing value at IPO.
Meta – News Corp	up to \$50M / yr	3-year term; chatbot responses + future training.
Amazon – New York Times	\$20–\$25M / yr	Amazon's first news deal; trains Nova + powers Alexa.
Dotdash Meredith – OpenAI	≥\$16M / yr	IAC filings; 40+ brands; also ad-targeting product.
OpenAI – Axel Springer	~\$13M / yr (3yr)	One-time historical dataset + ongoing licensing.
Financial Times – OpenAI	\$5–\$10M / yr	Non-exclusive; training + attributed display.
Wiley – (undisclosed)	\$23M + \$21M	Academic content, FY2024.
Shutterstock – Big Tech	\$25–\$50M / deal	Stock media; \$138M total AI-licensing revenue in 2024.
Stack Overflow – OpenAI/Google	undisclosed	59M+ Q&A pairs via OverflowAPI — the clearest code-adjacent licensing comp.

Pricing is migrating from flat annual fees toward usage-based and revenue-share structures (e.g., ProRata pays publishers 50% of downstream revenue). For code specifically, Stack Overflow is the closest licensed comp; a clean market for licensed source code barely exists — which is the opportunity.

4.3 Bottom-up: buyer spend and vendor-revenue proxies

Frontier infrastructure capex is enormous (Alphabet \$175–185B in 2026; Meta \$66–72B in 2025; Microsoft ~\$100B+ FY26), but the great majority is compute. Historically, data was a low-single-digit-to-~10% line item of training cost; however, it is the fastest-growing line — by some 2024–25 estimates human data now rivals or exceeds the marginal compute cost of frontier training. Even at a conservative 3–5% of the top labs' training/infra spend, the addressable data wallet exceeds \$10B and is rising.

A cleaner bottom-up read comes from managed-data vendors, whose revenue is (approximately) what labs pay for created data. Summing run-rates — Surge ~\$1.2B, Scale ~\$0.87–\$2B, Handshake ~\$1.1B, Mercor ~\$0.45B, Turing ~\$0.3B, Invisible ~\$0.13B — gives **~\$4–\$6B of expert/managed data spend alone**, on top of licensing deals and internal labeling. Combined with the ~\$3.9B datasets market and ~\$3.8B labeling market, the **total data-acquisition wallet is ~\$8–\$12B in 2026**, heading toward \$30B+ by the early 2030s. Enterprises separately spent ~\$37B on generative AI in 2025 (~\$18B on infrastructure/models/tooling; Menlo Ventures).

4.4 Seller supply pools today

Pool	Raw scale	Monetization today / liquidity
Legacy / non-moat code (D)	GitHub 630M repos; SWH 27B files	High latent supply, near-zero rights-clean liquidity. Only permissive corpora (The Stack ~6TB) are sale-ready — and legally contested.
Dead code (E)	1000s of defunct/pivoted cos/yr	Monetized only as patents via auctions (Nortel \$4.5B, Kodak \$525M). Source code almost never sold; brokered patent market ~\$290M/yr. Lowest liquidity, highest latent value.
Research-for-cash (F)	Millions of vetted experts	Active, professionalized, ~\$4–\$6B run-rate across Surge/Scale/Mercor/Turing/Handshake/Invisible; coding data at top of pay scale (\$85–\$125/hr).

5. Question 3 — How much could it be used?

A staged TAM / SAM / SOM for a code/data marketplace. Figures are scenario estimates with explicit assumptions, not forecasts; each step states its basis and confidence.

5.1 TAM — total addressable

The total spend on training data/code by model builders that could, in principle, transact through a marketplace.

- **Narrow lens (datasets only):** ~\$3.9B (2026) → ~\$16–\$23B by 2033–34 (GVR/FBI, 22–23% CAGR).
- **Broad lens (data-acquisition wallet = datasets + labeling + licensing + expert data):** ~\$8–\$12B (2026) → ~\$30B+ by the early 2030s.
- **Capex sanity check:** 3–5% of frontier training/infra spend implies a >\$10B data wallet today — consistent with the broad lens.

TAM anchor: ~\$8–\$12B in 2026, scaling to ~\$25–\$30B+ by the early 2030s.

5.2 SAM — serviceable (code-specific, rights-cleared)

No clean public figure exists for a “code data market,” so SAM is built from an explicit assumption stack:

Assumption	Estimate	Basis / confidence
Code as a share of the data-acquisition wallet	15–25%	Code is a core pretraining pillar (The Stack, StarCoder) + SWE-agent/RL boom + code-for-reasoning value. Assumption — med.
Implied code-data spend (2026)	\$1.5–\$3B	15–25% of \$8–\$12B broad wallet.
Rights-cleared / provenance-tracked subset	Greenfield	Almost no cleanly-licensed code corpora exist at scale today — Bitcode's wedge.
SAM growth rate	25–35% CAGR	Above-market, driven by agentic coding + RL environments + data-wall mitigations.
SAM (early 2030s)	\$6–\$12B+	Compounding the 2026 base at 25–35%.

5.3 SOM — serviceable obtainable (take-rate scenarios)

What Bitcode could realistically capture depends on (a) the GMV it intermediates and (b) its take rate, which is set by how much rights-clearing, curation and verification it owns. Marketplace comps bound the take rate: 0–3% passive (Databricks 0%, AWS 3%, Ocean ~0.3%) to ~30–40% curated (Mercor) and ~50% managed markup (Scale-style).

Model	Take rate	On \$150M GMV (early)	On \$1.5B GMV (scaled)
Passive listing venue	3%	~\$4.5M revenue	~\$45M revenue
Hybrid (some curation)	12–15%	~\$18–\$23M	~\$180–\$225M
Curated / rights-cleared / verified	30%	~\$45M	~\$450M

Interpretation. GMV figures are illustrative capture levels, not the full SAM: \$150M early GMV is a low-single-digit slice of a \$1.5–\$3B SAM; \$1.5B scaled GMV assumes Bitcode intermediates a meaningful fraction of a \$6–\$12B SAM. The realistic path runs from tens of millions of revenue in the early years (curated take on modest GMV) to \$100M–\$400M+ as GMV and share grow. The strategic choice — passive venue vs curated rights-clearing supplier — is the single biggest determinant of revenue, and the curated position is both more defensible and better matched to the legal-risk demand driver.

5.4 Growth drivers and sensitivity

- **Upside:** data-wall scarcity (2026–32), agentic-coding/RL-environment demand, tightening legal enforcement raising the value of clean rights, and shift to usage/rev-share pricing that rewards intermediaries.
- **Downside:** median-enterprise fine-tuning demand softening (a16z), synthetic data substituting for some human/code data, buyers vertically integrating supply (Meta–Scale), and thin take rates if Bitcode is perceived as a passive venue.

- **Sensitivity:** revenue scales roughly linearly with both take rate (3% → 30% is a 10× swing) and intermediated GMV share. The code-share-of-wallet assumption (15–25%) moves SAM by $\sim \pm 40\%$. These three levers dominate the model.

6. Competitive & comparable landscape

Bitcode sits between three comparable models: passive data marketplaces (low take, low curation), managed data vendors (high take, high curation), and IP exchanges (a cautionary tale).

Platform	Scale	Model	Take rate
Databricks Marketplace	Open (Delta Sharing)	Direct provider↔consumer	0%
AWS Data Exchange	300+ providers; 100+ PB	Cloud data subscriptions	3% (seller keeps 97%)
Ocean Protocol	Decentralized; modest volume	On-chain datatokens	~0.3% (config-dependent)
Snowflake Marketplace	1000s of listings	Usage/subscription	Not disclosed (private schedule)
Hugging Face Hub	\$13.5B val; 1M+ assets	Freemium/open-core	No data take (subs/compute)
Scale AI	\$29B val; \$0.87–\$2B rev	Vertically integrated data engine	~50%+ gross margin (markup)
Mercor	\$10B val; ~\$450M ARR	Expert-data / RL vendor	~30–40% (contractors keep 60–70%)
Surge AI	~\$1.2B rev; bootstrapped	RLHF / annotation	Markup-based (undisclosed)
IPXI (Ocean Tomo)	70+ members; closed 2015	Standardized IP-license units	Failed — cautionary comp
App Store / Steam (ref)	Dominant storefronts	Digital-goods distribution	~30% (reproducible-goods ceiling)

Positioning implication. Pure data marketplaces earn thin fees precisely because they add little beyond hosting. The defensible, high-take position is curated, rights-cleared, provenance-verified supply — benchmarked to Mercor (~30–40%) and the app-store ceiling (~30%) for reproducible digital goods. IPXI's failure warns that standardized IP-unit exchanges struggle without liquidity and buyer trust; Bitcode's answer is curation and a defensible rights trail, not standardization alone.

7. Risks, caveats & confidence

Risk / caveat	Detail	Confidence in data
Market-size definitional spread	“AI training data” varies 2–3× by definition; treat top-down figures as directional and cite the specific firm/definition.	High (well-documented)
Fine-tuning demand softening	a16z's CIO survey shows median-enterprise fine-tuning ROI declining as base models improve; durable demand is verticalized/code/RL data.	High (primary survey)
Supply vertical integration	Buyers may buy suppliers (Meta–Scale \$14.3B), shrinking the open market.	High
Take-rate compression	Passive data marketplaces price at 0–3%; a high take requires owning curation/rights-clearing. Scale's exact take is undisclosed.	Med
Synthetic-data substitution	Synthetic + verified code (e.g., NVIDIA Nemotron) can displace some human/licensed code demand.	Med
Legal complexity as double-edge	Rights-clearing is the wedge but also a cost/liability; provenance must be defensible (Copilot suit, GDPR on commit history).	High (litigation on record)
2026 valuation/ARR figures	Several lab valuations/ARR draw on secondary trackers (Sacra, getlatka); deal values and capex are primary-sourced.	Mixed (flagged inline)

8. Methodology & sources

Research fanned out across five angles (top-down sizing; frontier-lab buyers; fine-tuning buyers; seller-side IP/code monetization; code-data value + marketplace comps), fetching authoritative analyst, news, primary-filing and research sources. The 15 most load-bearing figures were adversarially re-verified by two independent fact-checkers seeking to refute each claim; 14 of 15 were confirmed against independent sources. The one correction — attributing the 22–24% CAGR to Grand View Research and Fortune Business Insights specifically (Precedence is ~16%) — is reflected above. Minor precision flags: News Corp “potentially” >\$250M incl. credits; Scale's exact take undisclosed; GitHub/Software Heritage figures are conservative floors (actuals higher).

Key sources

- Grand View Research — AI Training Dataset Market: grandviewresearch.com/industry-analysis/ai-training-dataset-market
- Fortune Business Insights — AI Training Dataset Market: fortunebusinessinsights.com/ai-training-dataset-market-109241
- Precedence Research — AI Training Dataset Market: precedenceresearch.com/ai-training-dataset-market
- Research and Markets — Custom LLM Training Platforms Outlook (2026): globenewswire.com/news-release/2026/04/20/3277235
- Grand View Research — Enterprise LLM Market: grandviewresearch.com/industry-analysis/enterprise-llm-market-report
- Quartz — The price of AI training data (\$5M to \$250M): qz.com/ai-training-data-pricing-licensing-deals-market-052126
- Variety / WSJ — News Corp–OpenAI licensing deal: variety.com/2024/digital/news/news-corp-openai-licensing-deal-1236013734
- TechCrunch — Reddit \$203M licensing disclosure: techcrunch.com/2024/02/22/reddit-says-its-made-203m-so-far-licensing-its-data
- Time / Forbes — Meta \$14.3B for 49% of Scale AI: time.com/7294699/meta-scale-ai-data-industry
- NPR / Authors Guild — Anthropic \$1.5B copyright settlement: npr.org/2025/09/05/nx-s1-5529404
- Epoch AI — Will we run out of data? epoch.ai/blog/will-we-run-out-of-data-limits-of-llm-scaling-based-on-human-generated-data
- Epoch AI — Trends in Frontier AI Model Count: arxiv.org/abs/2504.16138

- VentureBeat / Cohere — Code in pre-training improves non-coding tasks: venturebeat.com/ai/code-in-pre-training-data-improves-llms-performance-at-non-coding-tasks
- Hugging Face BigCode — StarCoder2 & The Stack v2: huggingface.co/blog/starcoder2
- GitHub Octoverse 2025: github.blog/news-insights/octoverse
- Software Heritage — archive as digital public good (2026): softwareheritage.org/2026/01/21
- GitHub Copilot Litigation: githubcopilotlitigation.com; The Register (2024) on DMCA dismissal
- Menlo Ventures — 2025 State of Generative AI in the Enterprise: menlovc.com/perspective/2025-the-state-of-generative-ai-in-the-enterprise
- a16z — How 100 Enterprise CIOs Are Building and Buying Gen AI (2025): a16z.com/ai-enterprise-2025
- Sacra — profiles: Scale AI, Surge AI, Mercor, Together AI, Hugging Face (secondary; directional)
- Marketplace take rates — Databricks (0%), AWS Data Exchange (3%), Ocean Protocol (~0.3%); Sharetribe/ MostlyMetrics comps